

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (EC) Examination

June 2014

EC201 Digital Electronics & Logic Design

Date: 03.06.2014, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Do as directed.

- (a) Convert $(19.75)_{10}$ into binary. [01]
- (b) Calculate: $(5531)_8 - (3261)_8 + (100)_{10} = (?)_{10}$ [02]
- (c) Perform subtractions of BCD number using 10's complement: $24 - 68 = ?$ [02]
- (d) Determine the base x : $(211)_x = (152)_8$ [02]

Q - 2 (a) Convert the following Boolean function into standard SOP and express it in terms of minterms. [05]

$$F(A,B,C) = AB + AC' + BC$$

- (b) Simplify the expression: $F(A,B,C) = (A' + B)(A + B + D)D'$ [05]
- (c) Draw the logic Diagram for following expression: [04]

$$F(A,B,C) = AB + BC(B+C)$$

OR

Q - 2 (a) Convert the following Boolean function into standard POS and express it in terms of minterms. [05]

$$F(A,B,C) = (A+B)(B+C')(A+C)$$

- (b) Simplify the expression: $F(A,B,C) = \sum(0,2,4,6)$ [05]
- (c) Draw the logic Diagram for following expression: [04]

$$F(A,B,C) = (A+B)'C'(C+D)$$

Q - 3 (a) Simplify following function using K-map [07]

$$F(w,x,y,z) = \sum(0,1,2,4,5,6,8,9,12,13,14)$$

(b) Simplify the following Boolean function by using the tabulation method: [07]

$$F(a,b,c,d) = \sum(0,1,2,8,10,11,14,15)$$

OR

Q - 3 (a) Simplify following function using K-map [07]

$$F = A'B'C' + B'CD' + A'BCD' + AB'C'$$

- (b) Determine the prime implicants of the function: [07]
$$F(w,x,y,z) = \sum(1,4,6,7,8,9,10,11,15)$$

SECTION – II

- Q - 4 (a)** Draw the Combinational logic diagram for Full Adder circuit with truth table and necessary functions. [04]
(b) Which two logic gates are known as Universal Gate? Justify answer with example. [03]
- Q - 5 (a)** Design BCD to Excess-3 code converter with Truth table and functions for 0 to 9 [07]
decimal numbers.
(b) Write a short note on 4 bit binary magnitude comparator with required diagram and [07]
equations.

OR

- Q - 5 (a)** Design BCD to decimal decoder. [07]
(b) Write a short note on multiplexer (4x1) & de-multiplexer (1x4) with Logic Diagrams. [07]
- Q - 6 (a)** Write characteristics truth tables and excitation tables for RS, JK, D and T flip-flops. [07]
(b) Design a sequential circuit with four flip-flops A,B,C and D. The next state of B, C [07]
and D are equal to the present state of A, B and C respectively. The next state of A is
equal to the OR logic of the present state of C and D.

OR

- Q - 6 (a)** Write a short note memory unit and explain read and write functions. [07]
(b) Design 4 bit BCD ripple counter. [07]
